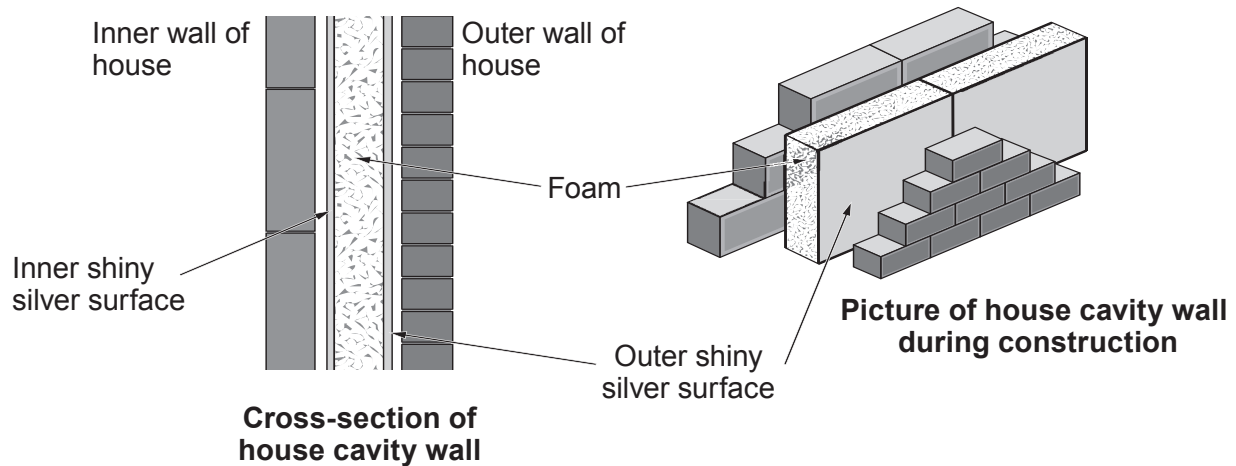


Examiner
only

- 3.** About 30 % of all the heat lost from an uninsulated house escapes through the walls.

Houses built in Wales since 1990 have wall insulation installed. One type of insulation that is commonly used is cavity board. This is a sheet of solid foam that is covered on both sides with shiny silver foil. During construction, sheets of cavity board are placed between the inner and outer walls of the house.



- (a) Explain why heat is lost through the walls of a house **and** how cavity board can reduce this heat loss by **conduction, convection and radiation**. [6 QER]



Examiner
only

(b) The table shows some information about three types of house with cavity board installed. All three houses use gas central heating.

Type of house	Cost of installing cavity board (£)	Saving per year (£)	Carbon footprint without cavity board installed (kg of CO ₂ /year)	CO ₂ saving with cavity board installed (kg/year)	Carbon footprint with cavity board installed (kg of CO ₂ /year)
detached	720	275	5 500	1 100	4 400
semi-detached	480	160	3 640		3 000
mid-terraced	385	110	2 440	440	

(i) The payback time of installing cavity board in a semi-detached house is 3 years. A semi-detached house owner claims that she has a longer payback time than the owner of the **mid-terraced** house. Explain whether you agree with her statement. *Space for workings.* [2]

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(ii) **Complete the table** for the two missing values. *Space for workings.* [2]

(iii) Use information in the table to explain which of the three types of house has the **smallest** impact on the environment. [2]

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